1 Introduction

1.1 Definition of masting

1. Observations of masting in relation to seed predation and flowering

1.2 Different hypotheses operating at different reproductive stages like filters

1. Pollination success (Pollination coupling)
2. Resources availability (Resource Matching)
3. Dispersal (Predation satiation)

1.3 Why traits can help link hypotheses to masting mechanisms

1. Mechanisms might not be mutually exclusive, but some might act stronger than the other for certain species
2. Traits thus can be important to understand masting through their relative importance

1.4 Trait-based perspective

1. Literature review on understanding masting from traits perspective
   (a) Wind pollination is related to higher CV
   (b) Animal dispersed seeds have higher CV than wind dispersed seeds
   (c) Larger seed is associate with higher CV
   (d) Recent papers also focus on resource perspective using leaf and shoot traits
2. Existing studies mainly trying to find relations between traits and masting.
3. Not clear connection between traits and hypotheses and potential evolutionary drivers

1.5 Introduce framework linking hypothesis x traits x masting

1. Linking traits to specific hypotheses helps us understand the underlying drivers of masting across different species
2. This framework is especially useful when considering whether evolutionary drivers are more similar within clades that are closely related

1.6 Use data from one specific geographical region

1. Here we use North American tree data to test this framework since they have information on their life-history
   (a) Masting info available
   (b) Includes both masting and non-masting species
   (c) Not using CV data because don't have stand-level traits data

2 Discussion

2.1 Summary of main findings

1. Pollination supports the pollination coupling hypothesis
2. Seed weight supports the predation satiation hypothesis
3. These act at the start and the end of the reproductive cycle, provide evolutionary incentive, which I will explain below
4. Discuss mechanisms in the context of existing literature
   (a) Previous literature on seed mass and masting think from the resource depletion perspective (Bigger seeds use up resources so cannot reproduce every year)
   (b) However, from evolutionary perspective, seeds attractive to animal dispersers will benefit more from masting
   (c) Relationship between masting and seed weight is stronger within biotic-dispersed group
5. Other traits show some pattern but statistically non-significant
6. Transition to alternative hypotheses

2.2 Masting is a population-level phenomenon

1. Other traits may matter to masting below the species level, like population level
2. Examples of species that mast only under certain environmental conditions (E.g., *Hymenaea coubaril*)
3. Traits need to be measured locally and match with masting data

2.3 Different species selected under different hypotheses

1. Introduce flowering masting vs. seed masting
2. Different species driven by different mechanisms
3. Need a more refined definition of masting and additional flowering-related traits
4. Transition to phylogeny

2.4 Phylogenetic signal for traits and masting

1. Many traits have strong phylogenetic signal, it is hard to tease out traits from masting, which limits the inference
2. Data is strongly nested in phylogeny, and data is sparse.

2.5 Deep evolutionary split between angiosperm and gymnosperm

1. Gymnosperms and angiosperms occupy fundamentally different functional spaces
   (a) More masting gymnosperm sp
   (b) Longer leaf longevity in gymnosperm
   (c) More monoecious species in gymnosperm
   (d) Wind pollination is universal among gymnosperms
   (e) Importance of looking for potential evolution support between gymnosperm and angiosperm
      i. Masting is more common in gymnosperm with specific functional traits
      ii. Test if masting was gradually diluted in angiosperm as they transitioned to biotic pollination and faster life histories

2.6 Importance of testing within clade

1. Existing literature points out the potentials within certain clade for evolutionary investigations (Satake & Kelly (2021))
2. This requires more data collection on species-level and population-level within clade, for example: (*Quercus.*)

References

Satake, A. & Kelly, D. (2021) Studying the genetic basis of masting. *Philosophical Transactions of the Royal Society B: Biological Sciences* **376**.